

环同态 (2)

例: 对 $\forall N, M \in \mathbb{Z}$ 且有 $N|M$. 定义映射 $p_N^M: \mathbb{Z}/M\mathbb{Z} \rightarrow \mathbb{Z}/N\mathbb{Z}$ 为:

$$p_N^M: \mathbb{Z}/M\mathbb{Z} \longrightarrow \mathbb{Z}/N\mathbb{Z}$$

$$[x]_M \longmapsto [x]_N$$

则有: p_N^M 是环同态.

p_{proof} : 对 $\forall [x]_M \in \mathbb{Z}/M\mathbb{Z}$ (其中 $x \in \mathbb{Z}$), 有:

$$\because x \in \mathbb{Z} \quad \therefore [x]_N \in \mathbb{Z}/N\mathbb{Z} \quad \therefore p_N^M([x]_M) = [x]_N \in \mathbb{Z}/N\mathbb{Z}$$

$$\therefore p_N^M(\mathbb{Z}/M\mathbb{Z}) \subseteq \mathbb{Z}/N\mathbb{Z}$$

对 $\forall [x]_M, [y]_M \in \mathbb{Z}/M\mathbb{Z}$ (其中 $x, y \in \mathbb{Z}$), 若 $[x]_M = [y]_M$, 则有:

$$x \equiv y \pmod{M} \quad \therefore M|x-y \quad \because N|M, M|x-y \quad \therefore N|x-y$$

$$\therefore x \equiv y \pmod{N} \quad \therefore [x]_N = [y]_N$$

$$\therefore p_N^M([x]_M) = [x]_N = [y]_N = p_N^M([y]_M)$$

$\therefore p_N^M: \mathbb{Z}/M\mathbb{Z} \rightarrow \mathbb{Z}/N\mathbb{Z}$ 是一个映射.

对 $\forall [x]_M, [y]_M \in \mathbb{Z}/M\mathbb{Z}$ (其中 $x, y \in \mathbb{Z}$), 有:

$$p_N^M([x]_M + [y]_M) = p_N^M([x+y]_M) = [x+y]_N = [x]_N + [y]_N$$

$$= p_N^M([x]_M) + p_N^M([y]_M)$$

$$p_N^M([x]_M \cdot [y]_M) = p_N^M([xy]_M) = [xy]_N = [x]_N \cdot [y]_N$$

$$= p_N^M([x]_M) \cdot p_N^M([y]_M)$$

$$\cancel{p_N^M([1]_N)} \quad p_N^M([1]_M) = [1]_N$$

$\therefore p_N^M: \mathbb{Z}/M\mathbb{Z} \rightarrow \mathbb{Z}/N\mathbb{Z}$ 是环同态. $\square$

例: ~~考虑环~~ $(R_i)_{i \in I}$ 是一族环, 则有: $(\prod_{i \in I} R_i, +, \cdot, (0_i)_{i \in I}, (1_i)_{i \in I})$ 是环.

对 $\forall i \in I$, 第 i 个投影映射 p_i 是:

$$p_i: \prod_{j \in I} R_j \longrightarrow R_i$$

$$(a_j)_{j \in I} \mapsto a_i$$

则有: p_i 是环同态.

Proof: 对 $\forall \cancel{(r_i)_{i \in I} \in \prod_{i \in I} R_i}$ $(r_j)_{j \in I} \in \prod_{j \in I} R_j$, $(r'_j)_{j \in I} \in \prod_{j \in I} R_j$, 有:

$$p_i((r_j)_{j \in I} + (r'_j)_{j \in I}) = p_i((r_j + r'_j)_{j \in I}) = r_i + r'_i = p_i((r_j)_{j \in I}) + p_i((r'_j)_{j \in I})$$

$$p_i((r_j)_{j \in I} \cdot (r'_j)_{j \in I}) = p_i((r_j \cdot r'_j)_{j \in I}) = r_i \cdot r'_i = p_i((r_j)_{j \in I}) \cdot p_i((r'_j)_{j \in I})$$

$$p_i((1_j)_{j \in I}) = 1_i \text{ 是环 } R_i \text{ 的么元.}$$

$\therefore p_i: \prod_{j \in I} R_j \longrightarrow R_i$ 是环同态. $\square$

Remark: 若对 $\forall j \in I$, $R_j = R$. 此时有 $\prod_{j \in I} R_j = R^I$.

$\therefore$ 对 $\forall i \in I$, 第 i 个投影映射 p_i 是:

$$\begin{aligned} \text{ev}_i &:= p_i: R^I \longrightarrow R_i = R \\ (\text{环同态}) \quad f &\longmapsto f(i) \end{aligned}$$

定义(环同构) R 和 R' 是环, $f: R \rightarrow R'$ 是环同态. 如果存在环同态 $g: R' \rightarrow R$ 使得 $gf = id_R$ 且 $fg = id_{R'}$, 则称 f 是环同构, 且 g 是 f 的逆. 此时我们也说环 R 和环 R' 同构.

Lemma: R 和 R' 是环, $f: R \rightarrow R'$ 是环同态, 则有: f 是双射 $\Leftrightarrow f$ 是环同构.

Proof: $(\Leftarrow): \because f$ 是环同构 $\therefore \exists$ 环同态 $g: R' \rightarrow R$ 使得 $gf = id_R$ 且 $fg = id_{R'}$

$\therefore f$ 是可逆映射 $\therefore f$ 是双射

$(\Rightarrow): \because f$ 是双射 $\therefore f$ 是可逆映射.

$\therefore$ 存在唯一的映射 $g: R' \rightarrow R$ 使得 $gf = id_R$ 且 $fg = id_{R'}$.

对 $\forall \lambda, \mu \in R'$, 有: $\because f: R \rightarrow R'$ 是满射

$\therefore \exists x \in R, \text{ s.t. } f(x) = \lambda. \quad \exists y \in R, \text{ s.t. } f(y) = \mu$

$\therefore g(\lambda + \mu) = g(f(x) + f(y)) = g(f(x+y)) = (gf)(x+y) = id_R(x+y)$

$= x+y = id_R(x) + id_R(y) = (gf)(x) + (gf)(y) = g(f(x)) + g(f(y))$

$= g(\lambda) + g(\mu)$

$g(\lambda \cdot \mu) = g(f(x) \cdot f(y)) = g(f(xy)) = (gf)(xy) = id_R(xy)$

$= xy = id_R(x) \cdot id_R(y) = (gf)(x) \cdot (gf)(y) = g(f(x)) \cdot g(f(y))$

$= g(\lambda) \cdot g(\mu)$

$\therefore f$ 是环同态 $\therefore f(1_R) = 1_{R'}$

$\therefore g(1_{R'}) = g(f(1_R)) = (gf)(1_R) = id_R(1_R) = 1_R \quad \therefore g$ 是环同态.

$\therefore f$ 是环同构. $\square$